Prof. Dr. Jonas Karthein

Assistant Professor
Cyclotron Institute / Dep. of Physics & Astronomy
Texas A&M University

karthein@tamu.edu
lab.karthein.com
ORCID 0000-0002-4306-9708
Curriculum vitae, November 2026

RESEARCH INTERESTS

We study the atomic nucleus as a window into physics beyond the Standard Model, combining precision laser spectroscopy, ion trapping and quantum sensing of short-lived isotopes and radioactive molecules. At the Texas A&M Cyclotron Institute we are building the Universal Action Spectrometer, a trap-based platform for background-free laser spectroscopy with single-ion sensitivity, to measure nuclear charge radii and electroweak, parity-violating nuclear properties such as the anapole moment. The lab leads an inaugural DOE Genesis Mission award on differentiable digital twins and AI-driven autonomous operation of precision experiments, runs collaborative experiments at FRIB, Argonne, Los Alamos and CERN/ISOLDE, and co-leads the NEPTUNE collaboration.

EDUCATION

2020	**Doctor rer. nat.**, Physics, Heidelberg University, Germany "Summa Cum Laude"; Advisor: Prof. Dr. Klaus Blaum Thesis: *Next-Generation Mass Spectrometry of Exotic Isotopes and Isomers*
2017	**M.Sc.**, Physics, Heidelberg University, Germany Thesis: *Precision mass measurements using the Phase-Imaging Ion-Cyclotron-Resonance detection technique*
2015	**B.Sc.**, Physics, Heidelberg University, Germany
2012	**Abitur**, Wendalinum Gymnasium, Germany

RESEARCH EXPERIENCE

2025 – now	**Assistant Professor**, Texas A&M University, USA
2021 – 2024	**Humboldt Postdoctoral Fellow**, Massachusetts Institute of Technology (MIT), USA
2017 – 2020	**CERN Doctoral Fellow**, CERN, Switzerland
2016 – 2017	**Master's Researcher**, CERN, Switzerland
2012 – 2016	**Bachelor's Researcher**, Max Planck Institute for Nuclear Physics, Germany

GRANTS & FUNDING

2026 – now	**DOE Genesis Mission (inaugural award)** (Lead PI), U.S. Department of Energy, Office of Science, award DE-SC0026653
2021 – 2025	**Humboldt Postdoctoral Fellowship** (Fellow), Alexander von Humboldt Foundation, Germany
2017 – 2020	**CERN Doctoral Scholarship (Gentner Program)** (Fellow), German Ministry for Education and Research

AWARDS & HONORS

2025	MIT Spot Award, MIT School of Science, USA
2023	FRIB Visiting Scholar Award, Facility for Rare Isotope Beams, USA
2022	Young Scientist Award, GSI/FAIR (GENCO), Germany
2022	Dissertation Award, European Physical Society, Spain
2021	MIT Spot Appreciation Award, MIT School of Science, USA
2021	Dissertation Award, German Physical Society, Germany
2020	Dissertation Award, Wilhelm und Else Heraeus Foundation, Germany
2018	Poster Presentation Award, Winter School on Trapped Charged Particles, France
2017	Poster Presentation Award, Euroschool on Exotic Beams, France

TEACHING

2025	PHYS 207 - Don't Panic #519-524: Electricity & Magnetism for Scientists and Engineers (97 students)
2025	PHYS 207 - Don't Panic #507-512: Electricity & Magnetism for Scientists and Engineers (114 students)
2021	Kaufman Teaching Certificate Program, MIT School of Science, USA
2021	MIT LEAPS Program Lecturer, MIT, USA
2018	Teaching Assistant, Heidelberg University, Germany

OUTREACH & SERVICE

Peer reviewer	Nature (Springer), European Physical Journal A (Springer), Nuclear Physics A (Elsevier), U.S. Department of Energy (DOE SCGSR)
2026 – now	Organizing Committee Leader, 2027 Exotic Beams Summer School, TAMU
2026	Outreach Lecture for Texas High School Physics Teachers (MIPEP), TAMU
2026	Outreach Event Demonstrations at the Texas A&M Physics and Engineering Festival, TAMU
2026	Outreach Lecture to High School Students, Saturday Morning Physics, TAMU
2025 – 2026	Organizing Committee Member, 2026 Exotic Beams Summer School, Oak Ridge National Lab
2025	Conference Organizing Committee Member, 2025 Low Energy Community Meeting, TAMU
2025	Sole Conference Organizer, Workshop on Precision Measurements with Radioisotopes, TAMU
2021 – 2024	Committee Member, MIT Radiation Protection Committee, USA
2021 – 2024	Environment, Health & Safety Officer, MIT School of Science, USA
2019 – 2023	Collaboration Website Design & Maintenance, www.cern.ch/isoltrap
2017 – 2020	Scientific Outreach, CERN Visitor Program
2014 – 2015	Conference Organizing Committee Leader, 2015 German Physical Society Spring Meeting

MENTORING & SUPERVISION

Postdoc research advisor

2. 2025 – now **Brooke Rickey**, TAMU, trap-based single-ion sensitive laser spectroscopy project
1. 2025 – now **Charlotte König**, TAMU & MIT, parity-violation and molecular break-up project

Graduate research advisor

7. 2025 – now **Savannah Limarenko**, TAMU, parity-violation and molecular break-up project
6. 2025 – now **Mason Moenter**, TAMU, trap-based single-ion sensitive laser spectroscopy project
5. 2025 – now **Nicholas Schnoor**, TAMU, trap-based single-ion sensitive laser spectroscopy project
4. 2023 – 2024 **Derick Emmanuel Gonzalez-Acevedo**, MIT, radio-frequency trap project
3. 2022 – 2024 **Haruka Kakioka**, MIT, radio-frequency trap and parity violation project
2. 2021 – 2024 **Scott Barrow Moroch**, MIT, parity violation project **Hertz Fellow 2022**
1. 2021 – 2024 **Silviu-Marian Udrescu**, MIT, parity violation project → now Assistant Professor, Princeton University (from January 2027)

Master's research advisor

2. 2025 **YeonJun Choi**, TAMU, molecular break-up project
1. 2017 – 2018 **Timo Steinsberger**, CERN, radio-frequency quadrupole trap optimization

Bachelor's and summer student research advisor

12. 2026 – now **Huy "Henry" Nguyen**, TAMU, ML simulation emulator
11. 2026 **Meg Gillies (REU)**, TAMU, CID gas cell
10. 2026 – now **Shreya Kulkarni**, TAMU, action spectrometer
9. 2025 – now **Riddhi Chhabra**, TAMU, wavelength meter stabilization
8. 2025 – now **Anna Goodson (TREND)**, TAMU, laser ion source development
7. 2025 – 2026 **Gabe Daniels (TREND)**, TAMU, Cyclotron Yield Chart
6. 2025 **Jack Thomas**, TAMU, 3D printing of electrodes
5. 2021 – 2024 **Andoni Fernandez Chiu**, MIT, hardware and control system
4. 2021 – 2022 **Juan Antonio Luera**, MIT, various sensors and control system
3. 2020 – 2021 **Parashar Mohapatra**, MIT (remote), ion-optical simulations → now Senior Electrical Engineer, Joby Aviation
2. 2020 – 2021 **Sarah Song**, CERN (remote), realistic buffer gas simulations

1. 2019 – 2020 **Adèle Perus**, CERN, ion-optical simulations and lab integration

THESIS COMMITTEES

2026 – now Ph.D. thesis committee member
2026 – now Master's thesis committee member
2025 Dissertation committee, University of Jyväskylä, Finland

BEAMTIME PROPOSALS

2025 **FRIB #25015**, oxygen isotopes for nuclear structure (co-author)
2025 **FRIB #25009**, zirconium isotopes for nuclear structure (co-author)
2025 **FRIB #25004**, francium isotopes for nuclear structure (co-author)
2023 **FRIB #23006**, nuclear astrophysics, r-process (spokesperson)
2023 **FRIB #23080**, precision nuclear structure (spokesperson)
2023 **FRIB #23060**, molecules for astrophysics (co-author)
2021 **FRIB #21023**, T-violation (spokesperson)
2021 **FRIB #21015**, precision nuclear structure (co-author)
2021 **INTC-P-612**, precision nuclear structure (co-author)
2017 **INTC-I-182**, Standard Model Tests (spokesperson, letter of intent)

PUBLICATIONS

26 peer-reviewed articles and preprints, 468 citations (INSPIRE-HEP, November 2026); ORCID 0000-0002-4306-9708

Pre-Peer-Review

26. CREMA: Collinear Resonance for Exotic Molecules and Atoms
Cruz FCP, Wilkins SG, Brinson AJ, Fischer P, Flayol M, ..., **Karthein J** et al.
arXiv:2610.30522
25. Nuclear charge radii of aluminum isotopes at the proton drip line
Brinson A, Rickey B, Arthuis P, Belley A, Campbell S, ..., **Karthein J** et al.
Under Review with Nature Physics. arXiv:2605.09139
** Postdoc Dr. Rickey led the experiment and analysis. The results were published in her dissertation.*
24. Hadronic parity violation: successes, challenges, and future prospects
Gardner S, **Karthein J**, Meißner UG, Muralidhara G, Navratil P, Snow WM.
Under Review with Elsevier for the Encyclopedia of Nuclear Physics. arXiv:2605.07826
** Invited review article.*

Peer-Reviewed

23. Hyperfine structures of 4s $^2S_{1/2}$, 5s $^2S_{1/2}$, 3p $^2P_{1/2,3/2}$, and mixed nd $^2D_{3/2,5/2}$ states in ^{27}Al I
Rickey BJ, Brinson A, Chakraborty A, Dockery A, Garcia Ruiz RF, **Karthein J** et al.
Phys. Rev. A 113, 042824 (2026). doi
** Postdoc Dr. Rickey led the experiment, analysis, and publication.*
22. Precision mass measurement of ^{61}Zn and updated Ft value for conserved vector current tests
Giesel PF, Atanasov D, Blaum K, Fischer P, **Karthein J** et al.
Phys. Rev. C 113, 034318 (2026) doi
21. Commissioning the Resonant Ionization Spectroscopy Experiment at FRIB
Brinson AJ, Rickey BJ, Allmond JM, Dockery A, Chiu AF, ..., **Karthein J** et al.
Phys. Rev. Res. 8, 013276 (2026) arXiv:2511.08881 doi
20. Charge radii measurements of exotic tin isotopes in the proximity of N = 50 and N = 82
Gustafsson FP, Rodríguez LV, Garcia Ruiz RF, Miyagi T, Bai SW, ..., **Karthein J** et al.
Phys. Rev. Lett. 135, 222501 (2025) arXiv:2504.17060 doi
19. Extending the low-Z "border" of the A=100 region of deformation with precision mass spectrometry of $^{96-98}$Kr
Lunney D, Manea V, Mougeot M, Nies L, Althubiti NA, ..., **Karthein J** et al.
Phys. Rev. C 112, 034307 (2025) arXiv:2509.05538 doi
18. Variations in the charge radii of indium isotopes between N = 52 and 82
Vernon AR, Binnersley CL, Sahoo BK, Billowes J, Bissell ML, ..., **Karthein J** et al.
Phys. Rev. C 111, 064325 (2025) **Editor's Suggestion** doi

17. Electromagnetic properties of indium isotopes elucidate the doubly magic character of ^{100}Sn
Karthein J, Ricketts CM, Garcia Ruiz RF, Billowes J, Binnersley CL et al.
Nature Phys. 20, 1719 (2024) arXiv:2310.15093 doi
16. Electroweak Nuclear Properties from Single Molecular Ions in a Penning Trap
Karthein J, Udrescu SM, Moroch SB, Belosevic I, Blaum K et al.
Phys. Rev. Lett. 133, 033003 (2024) arXiv:2310.11192 doi
15. Nuclear charge radii of silicon isotopes
König K, Berengut JC, Borschevsky A, Brinson A, Brown BA, ..., **Karthein J** et al.
Phys. Rev. Lett. 132, 162502 (2024) arXiv:2309.02037 doi
14. Opportunities for Fundamental Physics Research with Radioactive Molecules
Arrowsmith-Kron G, Athanasakis-Kaklamanakis M, Au M, Ballof J, Berger R, ..., **Karthein J** et al.
Rep. Prog. Phys. 87, 084301 (2024) arXiv:2302.02165 doi
13. Isomeric Excitation Energy for 99Inm from Mass Spectrometry Reveals Constant Trend Next to Doubly Magic ^{100}Sn
Nies L, Atanasov D, Athanasakis-Kaklamanakis M, Au M, Blaum K, ..., **Karthein J** et al.
Phys. Rev. Lett. 131, 022502 (2023) arXiv:2306.02033 doi
12. KATRIN background due to surface radioimpurities
Fränkle F, Schaller A, Blaum K, Otten E, Bornschein L, ..., **Karthein J** et al.
Astropart. Phys. 138, 102686 (2022) arXiv:2011.05107 doi
11. Mass measurements of $^{99-101}$In challenge ab initio nuclear theory of the nuclide ^{100}Sn
Mougeot M, Atanasov D, **Karthein J**, Wolf RN, Ascher P et al.
Nature Phys. 17, 1099 (2021) arXiv:2109.10673 doi
** MM, DA, and JK share first authorship for equal contributions.*
10. Analysis methods and code for very high-precision mass measurements of unstable isotopes
Karthein J, Atanasov D, Blaum K, Lunney D, Manea V, Mougeot M.
Comp. Phys. Comm. 267, 108070 (2021) arXiv:2102.10413 doi
9. Masses of short-lived ^{49}Sc, ^{50}Sc, ^{70}As, ^{73}Br, and stable ^{196}Hg nuclides
Kulikov I, Algora A, Atanasov D, Ascher P, Blaum K, ..., **Karthein J** et al.
Nucl. Phys. A 1002, 121990 (2020) arXiv:2010.15467 doi
8. Examining the N=28 shell closure through high-precision mass measurements of $^{46-48}$Ar
Mougeot M, Atanasov D, Barbieri C, Blaum K, Breitenfeld M, ..., **Karthein J** et al.
Phys. Rev. C 102, 014301 (2020) arXiv:2006.02712 doi
7. First glimpse of the N=82 shell closure below Z=50 from masses of neutron-rich cadmium isotopes and isomers
Manea V, **Karthein J**, Atanasov D, Bender M, Blaum K et al.
Phys. Rev. Lett. 124, 092502 (2020) arXiv:2001.05075 doi
** VM and JK share first authorship for equal contributions.*
6. Improved stability of multi-reflection time-of-flight mass spectrometers through passive and active voltage stabilization
Wienholtz F, Blaum K, **Karthein J**, Lunney D, Malbrunot-Ettenauer S et al.
Nucl. Inst. Meth. Phys. B 463, 348 (2020). doi
5. QEC-value determination for ^{21}Na$\rightarrow^{21}$Ne and ^{23}Mg$\rightarrow^{23}$Na mirror-nuclei decays using high-precision mass spectrometry with ISOLTRAP at ISOLDE/CERN
Karthein J, Atanasov D, Blaum K, Breitenfeldt M, Bondar V et al.
Phys. Rev. C 100, 015502 (2019) arXiv:1906.01538 doi
4. Direct decay-energy measurement as a route to the neutrino mass
Karthein J, Atanasov D, Blaum K, Eliseev S, Filianin P et al.
Hyperfine Int. 240, 61 (2019) arXiv:1905.05510 doi
3. Long-term monitoring of the internal energy distribution of isolated cluster systems
Breitenfeldt C, Blaum K, George S, Göck J, Guzmán-Ramírez G, **Karthein J**, et al.
Phys. Rev. Lett. 120, 253001 (2018). arXiv:1805.11403 doi
2. Photodissociation of an Internally Cold Beam of CH+ ions in a Cryogenic Storage Ring
O'connor AP, Becker A, Blaum K, Breitenfeldt C, George S, ..., **Karthein J**, et al.
Phys. Rev. Lett. 116, 113002 (2016). doi
1. The Cryogenic Storage Ring CSR
von Hahn R, Becker A, Berg F, Blaum K, Breitenfeldt C, ..., **Karthein J** et al.
Rev. Sc. Instr. 87, 063115 (2016) arXiv:1606.01525 doi

Theses

2. Next-Generation Mass Spectrometry of Exotic Isotopes and Isomers
   Karthein J.
   Heidelberg University (2020) doi
1. Precision mass measurements using the Phase-Imaging Ion-Cyclotron-Resonance detection technique
   Karthein J.
   Heidelberg University (2017) doi

Other

6. Karthein Lab document templates and style: Typst themes (slides, posters, lecture notes, CV, lab notebook), palette, matplotlib style, guides
   Karthein J.
   GitHub, jonas-ka/lab-templates (2026). repository
5. Reproducible research talks with Typst + Touying, a curated figure library and Claude Code: pipeline, themes, style, recipe (template)
   Karthein J.
   GitHub, jonas-ka/talks-repo-template (2026). repository
4. Karthein Lab Website
   Karthein J.
   GitHub, jonas-ka/karthein-lab-website (2026). repository
3. PI-ICR analysis software
   Karthein J.
   GitHub, Zenodo (2020). repository
2. Python script to produce publication-ready nuclear chart plots
   Karthein J.
   GitHub, jonas-ka/nuclear-chart-plotter (2019). repository
1. MRToF analysis software
   Karthein J.
   GitHub (2019). repository

PRESENTATIONS

64 talks and posters, 33 invited

64. 2026/12 **Nuclear Seminar, University of Kentucky** **invited**, "Trapped Radioactive Molecules: A New Path for Hadronic Parity Violation Measurements," Lexington, KY, USA *(scheduled)*
63. 2026/11 **DOE Genesis Mission HENP Community Meeting** **lightning talk**, "Scalable Agentic Digital Twins for Autonomous Precision Facilities," Washington, DC, USA
62. 2026/11 **TRIUMF Colloquium**, "Agentic Digital Twins and Radioactive Molecules at the Texas A&M Cyclotron Institute," Vancouver, BC, Canada
61. 2026/11 **MUI-QC Workshop** **invited**, "Agentic Digital Twins for Self-Driving Precision Experiments," College Station, TX, USA
60. 2026/10 **Texas A&M University System Artificial Intelligence Research & Innovation Meeting**, "Autonomous Nuclear Experiments: Agentic Digital Twins supported by the DOE Genesis Mission," Kyle Field, Texas A&M University, College Station, TX, USA
59. 2026/09 **AWS**, "Digital Twins for Self-Driving Physics Experiments," virtual
58. 2026/09 **NVIDIA VISION**, "Fast, Differentiable Simulations for Self-Driving Physics Experiments," College Station, TX, USA
57. 2026/08 **RadIs Workshop** **invited**, "Isotope Opportunities at Texas A&M University," Oak Ridge National Laboratory, TN, USA
56. 2026/08 **NAMO Workshop** **invited**, "Agentic Digital Twins for Autonomous Nuclear Precision Facilities," Argonne National Laboratory, IL, USA
55. 2026/08 **AI for Nuclear Summit** **invited**, "Agentic Digital Twins for Autonomous Nuclear Precision Facilities," College Station, TX, USA
54. 2026/08 **Cyclotron Seminar**, "Artificial Intelligence and Machine Learning for Nuclear Physics," College Station, TX, USA
53. 2026/06 **Mitchell Institute Physics Enhancement Program (MIPEP)**, "Introduction to Nuclear Physics," College Station, TX, USA

52. 2026/04 **Ariel Science Workshop 2026** Invited Panel, “Experimental horizons for BSM and electroweak interactions using AMO techniques and rare isotope beams,” TRIUMF, Vancouver, BC, Canada
51. 2026/03 **Cyclotron Seminar** invited, “The mechanics of money - Personal Finance for Physicists,” College Station, TX, USA
50. 2026/03 **APS Global** invited, “Trapped (Radioactive) Molecules -- A New Path for Laser-based Parity Violation Measurements,” Denver, CO, USA
49. 2026/02 **Saturday Morning Physics Lecture** invited, “Atoms and Molecules as Windows into the Atomic Nucleus,” College Station, TX, USA
48. 2025/10 **Seminar Talk IQSE** invited, “Quantum Sensing of Nuclei,” College Station, TX, USA
47. 2025/08 **ArtSci Quantum Research Meeting**, “Quantum Sensing of Nuclei at NEPTUNE,” College Station, TX, USA
46. 2025/08 **Low-Energy Community Meeting**, “Summary and Findings of Workshop on Precision Measurements of Radioisotopes (Aug. 11-12),” College Station, TX, USA
45. 2025/08 **Workshop on Precision Measurement with Radioisotopes**, “Nuclear Structure Studies at NEPTUNE and Texas A&M Cyclotron Institute,” College Station, TX, USA
44. 2025/01 **Joint TAMU-UT Quantum Workshop**, “Quantum Sensing with Nuclei,” College Station, TX, USA
43. 2024/10 **APS DNP 2024**, “Shedding Light on the Doubly Magic Character of 100Sn (N=Z=50),” Boston, MA, USA
42. 2024/06 **PLATAN24**, “Parity-Violating Nuclear Effects From Single Molecules In A Penning Trap with NEPTUNE,” Jyväskylä, Finland
41. 2024/03 **Texas A&M Physics Colloquium** invited colloquium, “Revealing Unknown Nuclear Properties with Next-Gen Precision Techniques,” College Station, TX, USA
40. 2024/02 **Colorado School of Mines Physics Colloquium** invited colloquium, “Revealing Unknown Nuclear Properties with Next-Gen Precision Techniques,” Golden, CO, USA
39. 2024/02 **LBNL Nuclear Physics Seminar** invited seminar, “Revealing Unknown Nuclear Properties with Next-Gen Precision Techniques,” Berkeley, CA, USA
38. 2023/08 **Argonne Physics Seminar** invited seminar, “Precision Techniques for Pioneering Symmetry and Structure Studies,” Lemont, IL, USA
37. 2023/08 **GAIN23** invited, “Laser Spectroscopy on Radioactive Molecules,” Boston, MA, USA
36. 2023/02 **Winter Workshop on Nuclear Dynamics** invited, “Precision Nuclear Experiments using Ion Traps and Lasers at FRIB,” Puerto Vallarta, Mexico
35. 2023/01 **FRIB Seminar** invited, “Fundamental Symmetries from Precision Experiments using Ion Traps and Lasers,” East Lansing, MI, USA
34. 2022/10 **US Long-Range-Plan Workshop for Fundamental Symmetries** invited, “Symmetry-violating measurements using single molecular ions in a Penning Trap,” East Lansing, USA (virtual)
33. 2022/10 **Department of Energy Institutional Review at MIT**, “Single Trapped Molecular Ions as Quantum Sensors for Symmetry-Violating Nuclear Properties,” Cambridge, MA, USA
32. 2022/10 **EuNPC22** invited symposium, award, “High-precision measurements in the direct vicinity of the doubly magic 100Sn(N=Z=50) at ISOLDE/CERN,” Santiago de Compostella, Spain
31. 2022/09 **Trapped Charged Particles (TCP22)**, “Symmetry-violating nuclear properties from single molecular ions in a Penning trap,” Glashütten, Germany
30. 2022/09 **NuSTAR Conference** invited symposium, award, “High-precision mass measurements in the direct vicinity of the doubly magic 100Sn(N=Z=50) at ISOLDE/CERN,” Darmstadt, Germany
29. 2022/07 **MRTOF Workshop**, “Exotic Molecules and Atoms Lab at MIT,” Darmstadt, Germany
28. 2022/03 **Spring conference of the German Physical Society** invited symposium, “High-precision mass measurements in the direct vicinity of the doubly magic 100Sn(N=Z=50) at ISOLDE/CERN,” virtual
27. 2022/01 **Argonne Physics Seminar** invited seminar, “High-precision mass measurements in the direct vicinity of the doubly magic 100Sn(N=Z=50) at ISOLDE/CERN,” virtual
26. 2022/01 **INT Parity Nonconservation II** invited, “Towards measurements of symmetry-violating nuclear properties using single molecular ions in a Penning trap,” Seattle, WA, USA
25. 2021/11 **Humboldt Networking Conference**, “Parity Violation from Single Molecular Ions in a Penning trap,” virtual
24. 2021/10 **DNP Conference of the American Physical Society** invited, “Towards measurements of symmetry-violating nuclear properties using single molecular ions in a Penning trap,” virtual

23. 2021/08 **Dissertation award symposium** invited symposium, award, “Next-Generation Mass Spectroscopy on Exotic Isotopes and Isomers,” German Physical Society (DPG-SMuK), virtual
22. 2021/08 **Spring conference of the German Physical Society (DPG21)**, “High-precision mass measurements in the direct vicinity of the doubly magic 100Sn(N=Z=50) at ISOLDE/CERN,” virtual
21. 2021/06 **Opportunities with Radioactive Molecules** invited, “Symmetry-violating properties from single molecular ions in a Penning trap,” Massachusetts Institute of Technology, Cambridge, MA, USA
20. 2021/01 **CRIS future workshop**, “Precision Spectroscopy of Single Molecular Ions in a Penning Trap,” virtual
19. 2020/12 **Dissertation award symposium** invited symposium, award, “Next-Generation Mass Spectroscopy on Exotic Isotopes and Isomers,” Heidelberg University, Germany
18. 2020/09 **MIT Laboratory for Nuclear Science Seminar** invited seminar, “High-Precision Mass Measurements of Radionuclides and Their Applications,” Massachusetts Institute of Technology, Cambridge, MA, USA
17. 2019/12 **LBNL Nuclear Physics Forum** invited seminar, “High-Precision Mass Measurements for Nuclear Structure and Astrophysics Studies,” Lawrence Berkeley National Laboratory, Berkeley, CA, USA
16. 2019/09 **MATS workshop**, “ISOLTRAP status,” Oberwesel, Germany
15. 2019/07 **PI-ICR workshop**, “ISOLTRAP’s PI-ICR analysis software,” MPIK, Heidelberg, Germany
14. 2019/06 **Gordon Research Conference** invited, “Precision Mass Measurements of Mirror Nuclei,” Colby-Sawyer College, New London, NH, USA
13. 2019/06 **National Nuclear Physics Summer School**, “Phase-imaging ion-cyclotron-resonance detection with ISOLTRAP at CERN,” Yale University, New Haven, CT, USA
12. 2019/05 **FAIRness19**, “Direct decay-energy measurement as a route to the neutrino mass,” Genova, Italy
11. 2019/03 **ISOLDE seminar** invited seminar, “Probing nuclear isomers using phase-imaging ion-cyclotron-resonance detection with ISOLTRAP at CERN,” CERN, Geneva, Switzerland
10. 2019/02 **Arbeitstreffen Kernphysik** invited, “High-precision mass measurements for nuclear astrophysics studies using ISOLTRAP/CERN,” Schleching, Germany
9. 2018/09 **Trapped Charged Particles (TCP18)**, “Probing nuclear isomers using phase-imaging ion-cyclotron-resonance detection with ISOLTRAP at CERN,” Traverse City, MI, USA
8. 2018/09 **Measurements with an Advanced Trapping System (MATS)**, “Probing nuclear isomers using phase-imaging ion-cyclotron-resonance detection with ISOLTRAP/CERN,” Hermeskeil, Germany
7. 2018/02 **Spring conference of the German Physical Society (DPG18)**, “Phase-imaging ion-cyclotron-resonance detection with ISOLTRAP at CERN,” Bochum, Germany
6. 2018/01 **Winter School on Physics with Trapped Charged Particles** award, “The phase-imaging ion-cyclotron-resonance detection technique used at ISOLTRAP/CERN,” Les Houches, France
5. 2017/12 **ISOLDE workshop**, “The phase-imaging ion-cyclotron-resonance detection technique used at ISOLTRAP/CERN,” CERN, Geneva, Switzerland
4. 2017/08 **Euroschool on Exotic Beams** award, “Implementation of the phase-imaging ion-cyclotron-resonance detection technique at ISOLTRAP/CERN,” Cabourg, France
3. 2017/04 **MPIK Institutional review**, “Precision measurements of nuclear ground-state properties with lasers and traps,” Max Planck Institute for Nuclear Physics, Heidelberg, Germany
2. 2017/03 **Spring conference of the German Physical Society**, “Implementation of the phase-imaging ion-cyclotron-resonance detection technique at ISOLTRAP/CERN,” Mainz, Germany
1. 2016/05 **Solid-state Master’s seminar**, “Order parameter in (un)conventional superconductors,” Heidelberg University, Germany